

LET'S PLAY TOGETHER

POWER vs PIGGY

START

MENU

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CONTENTS

PART 2 : FORZEN LAKE

ALGORITHM & OPTIMIZATION

PART 3 : ARCHER VS PIGGY

OOP DESIGN

RESULTS & DEMO

The background is a pixel art scene. At the top, a bright yellow rectangular sign with rounded corners and a dark blue border is suspended by two dark blue vertical lines. The sign contains the text 'FORZEN LAKE' in a dark blue, pixelated, sans-serif font. Below this, a white rectangular box with a dark blue border contains the text 'PART 2' in the same font. In the foreground, a grey chain-link fence runs horizontally across the middle. Below the fence is a light blue, pixelated ground surface with several small green patches. Four penguins are depicted in a pixel art style: one is flying in the upper left sky, another is flying in the upper right sky, one is standing on the ground to the left of the fence, and one is standing on the ground to the right of the fence. The penguins have grey bodies, white bellies, and small red and black details on their heads.

FORZEN LAKE

PART 2



METHOD

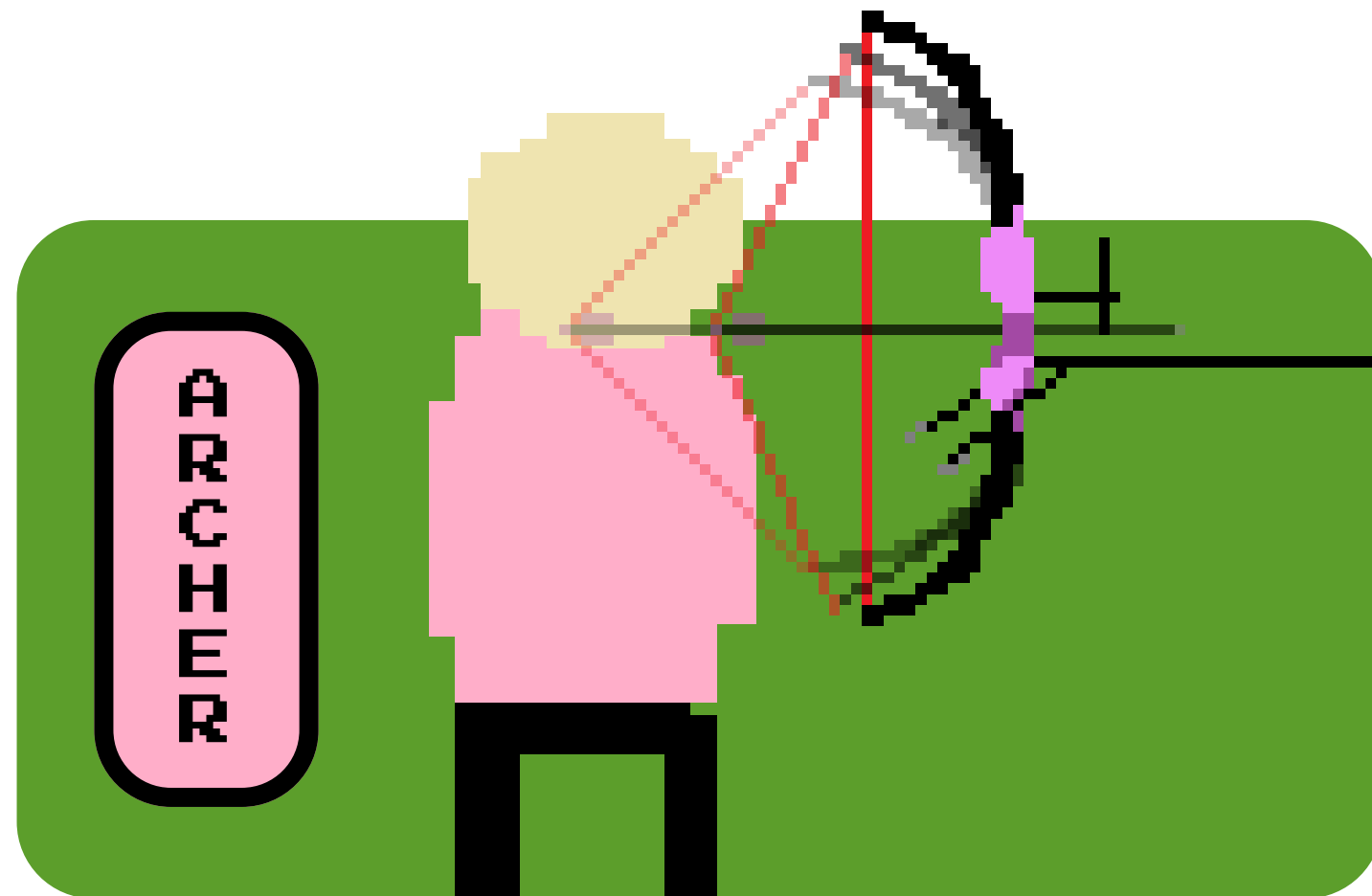
1. *Optimistic Initialization*
2. *Epsilon-Greedy Exploration with Decay*
3. *Learning Rate Decay*
4. *Reward Shaping*
5. *High Discount Factor*



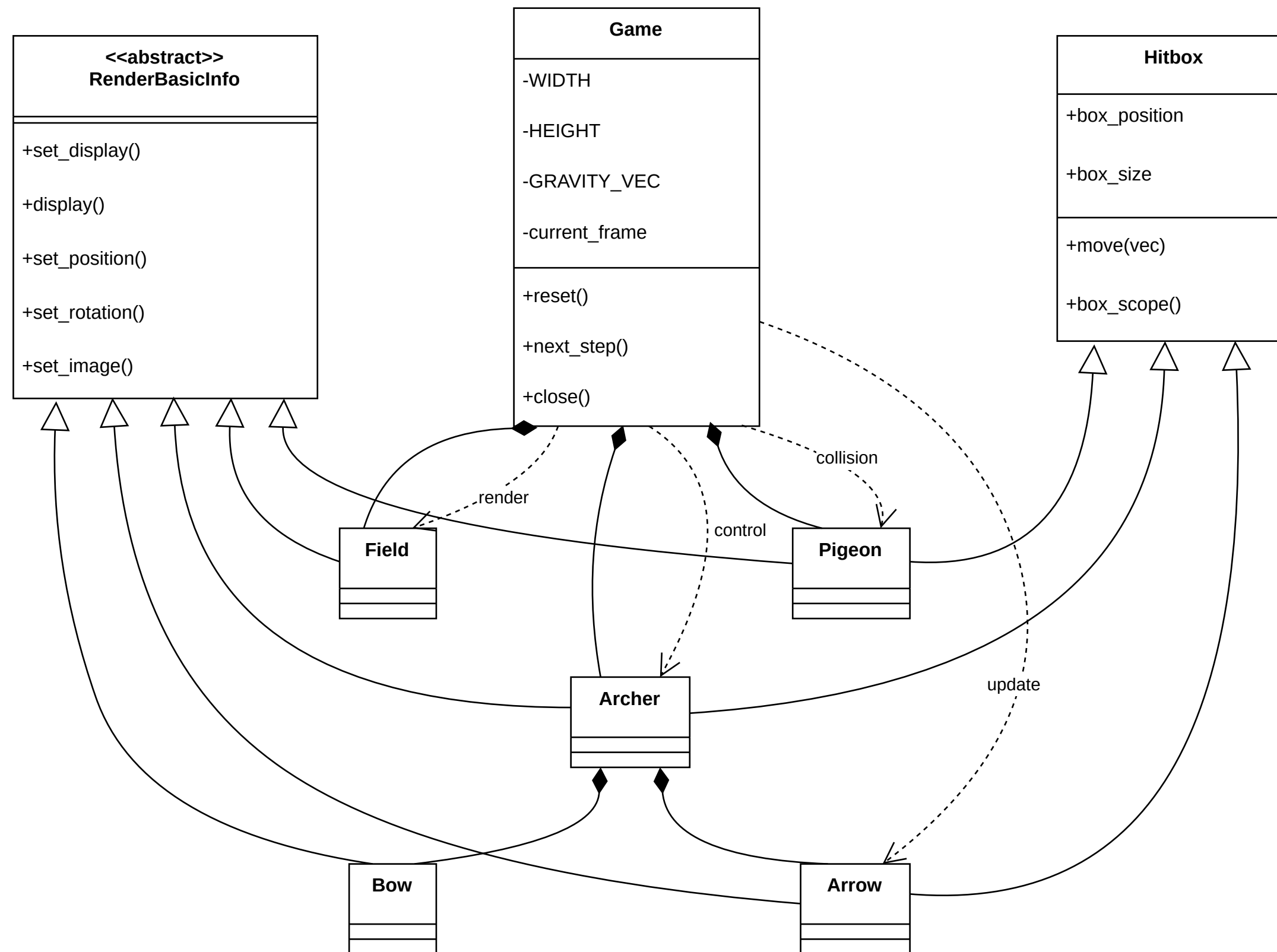
ARCHER VS PIGGY

PART 3

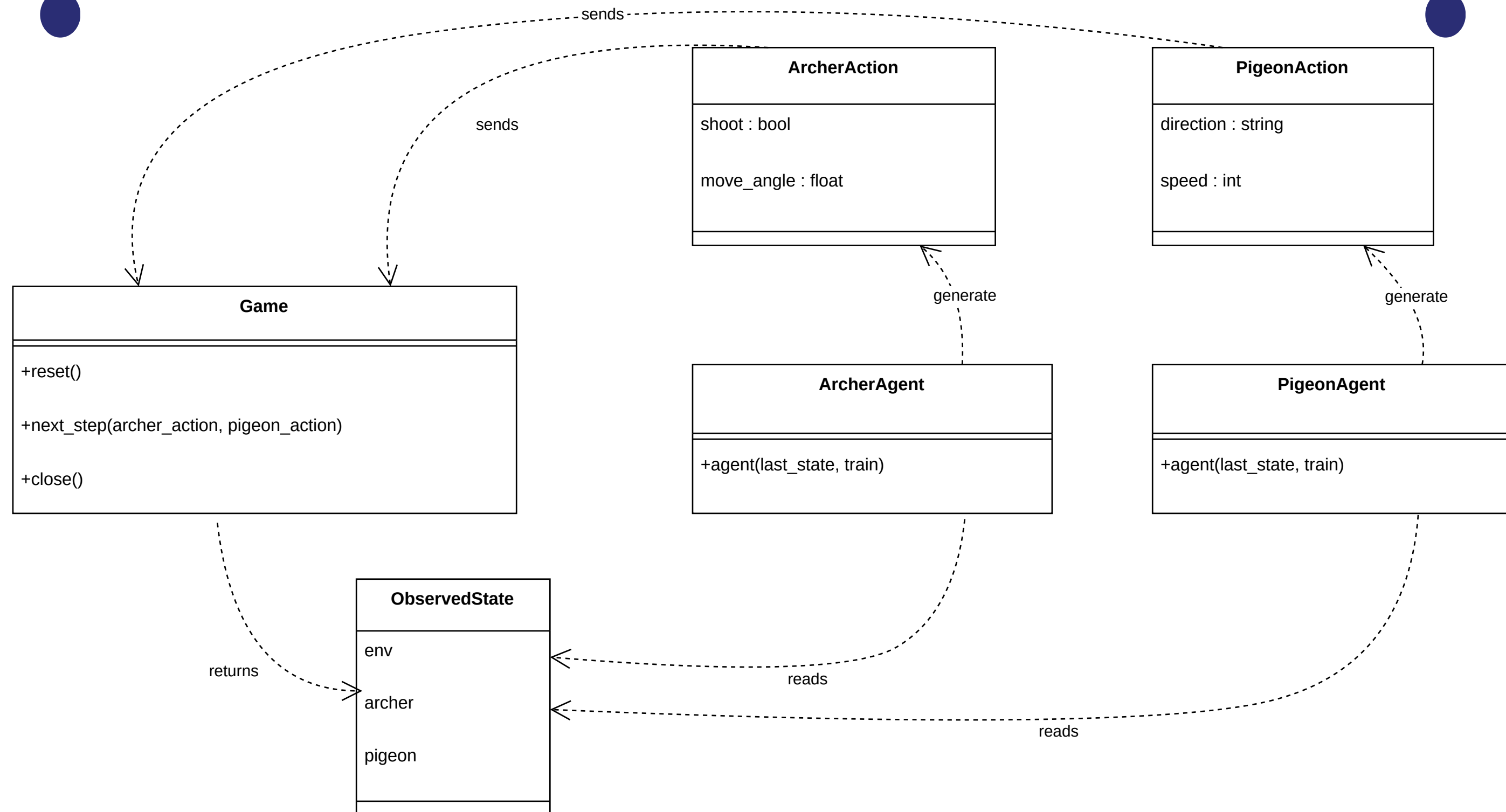
CHARACTER



METHOD



METHOD



DEMO

```
File Edit Selection View Go Run ... 114-1-OOP-group-project
```

EXPLORER

- 114-1-OOP-GROUP-PROJE...
- part2
- part3
 - __pycache__
 - Archer_Pippy.cpyth...
 - game_env.cpython-...
 - oop_project_env.cp...
 - warehouse_robot.c...
 - dataset
 - sprites
 - Archer_Pippy.py
 - game_env.py
 - main.py
 - oop_project_env.py
 - warehouse_robot.py
- project_requirement
- OOP_Final_Presentati...
- Project.pdf
- REPORT
 - README.md
 - .gitignore
 - NOTE.md
 - README.md
- OUTLINE
- TIMELINE

main.py

```
part3 > main.py > run
349 def run(episodes=1, render=True, archer_train=False, pigeon_train=False, pigeon_mode="random_agent"):
397     print(f"Final Safe Y Range: [{pigeonAgent.safe_y_min:.1f}, {pigeonAgent.safe_y_max:.1f}]")
398
399     # 測試階段統計
400     if not archer_train and not pigeon_train:
401         print(f"\n=== Test Results ===")
402         print(f"Total Episodes: {episodes}")
403         print(f"Archer Wins: {archer_wins} ({archer_wins/episodes*100:.1f}%)")
404         print(f"Pigeon Wins: {pigeon_wins} ({pigeon_wins/episodes*100:.1f}%)")
405         print(f"Win Rate - Archer: {archer_wins}/{episodes}, Pigeon: {pigeon_wins}/{episodes}")
406
407     game.close()
408
409
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS TEROSHDL: LOG REPORT TEROSHDL: TIMING

開始訓練

【階段 2】訓練 Pigeon 對抗已訓練好的 Archer...

[Archer] 載入訓練參數: Height Adjustment = 5.57

Episode 10: Pigeon Success Rate (last 7): 100.0%, Safe Y Range = [229.4, 378.4]

Episode 20: Pigeon Success Rate (last 10): 100.0%, Safe Y Range = [229.4, 378.4]

Episode 30: Pigeon Success Rate (last 10): 100.0%, Safe Y Range = [229.4, 378.4]

Episode 40: Pigeon Success Rate (last 10): 100.0%, Safe Y Range = [229.4, 378.4]

Episode 50: Pigeon Success Rate (last 10): 100.0%, Safe Y Range = [229.4, 378.4]

Episode 60: Pigeon Success Rate (last 10): 100.0%, Safe Y Range = [229.4, 378.4]

Episode 70: Pigeon Success Rate (last 10): 100.0%, Safe Y Range = [229.4, 378.4]

Episode 80: Pigeon Success Rate (last 10): 100.0%, Safe Y Range = [229.4, 378.4]

Ln 356, Col 20 Spaces: 4 UTF-8 CRLF {} Python .venv (3.11.9)





YEAY! GAME IS OVER!



THANK YOU

STOP

